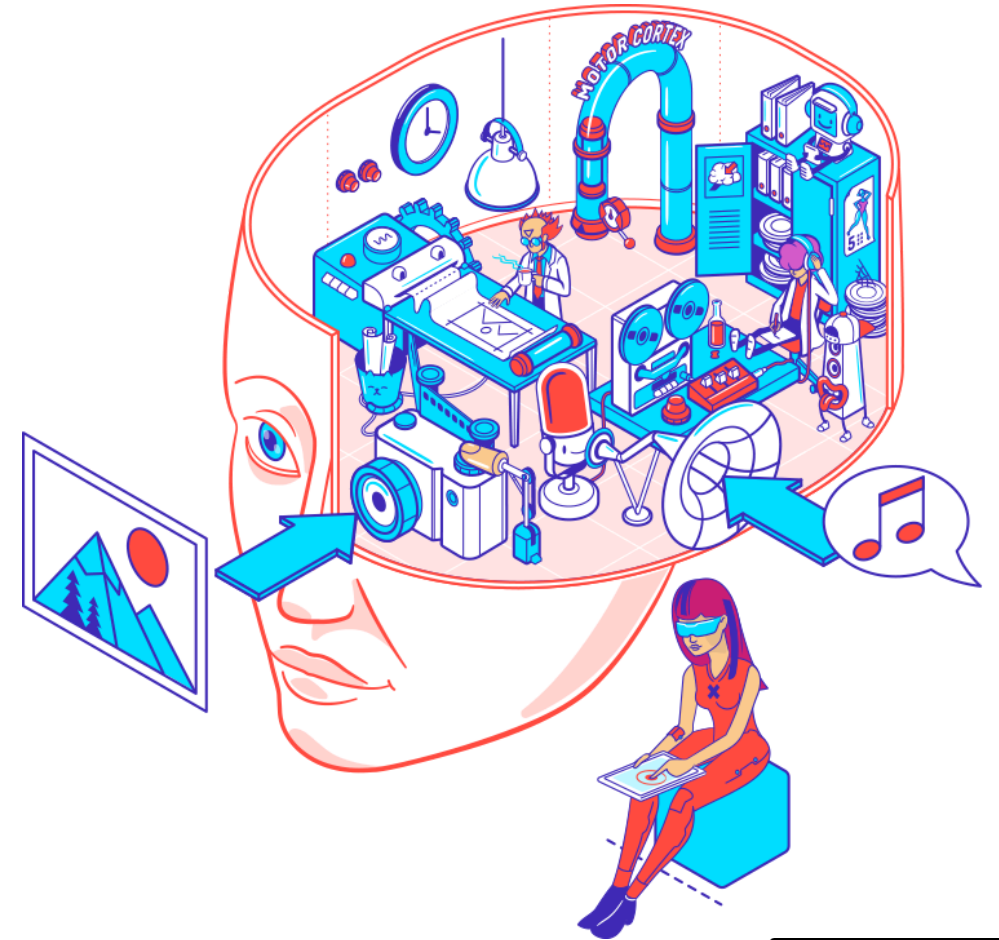


MEDIA PROGRAMMING

UNIT 1: HUMAN PERCEPTION

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Faculty of Design, Media and Information
Hamburg University of Applied Sciences



LEARNING OUTCOME

GRAPHICS PROGRAMMING

UNIT 1: HUMAN PERCEPTION

INTRO

- Understanding **human information processing**
- Knowledge about **visual perception, light, color**
- Knowing selection & filtering of information, multi-sensory integration

INFORMATION PROCESSING

GRAPHICS PROGRAMMING

UNIT 1: HUMAN PERCEPTION

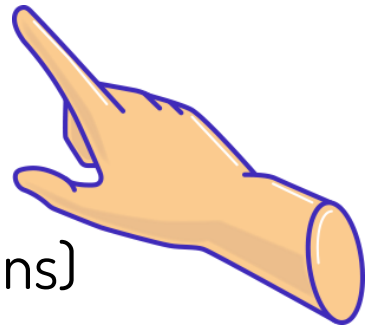
INTRO



Perception
(Receptors)

Cognition
(Central nervous system)

Action
(Motor neurons)



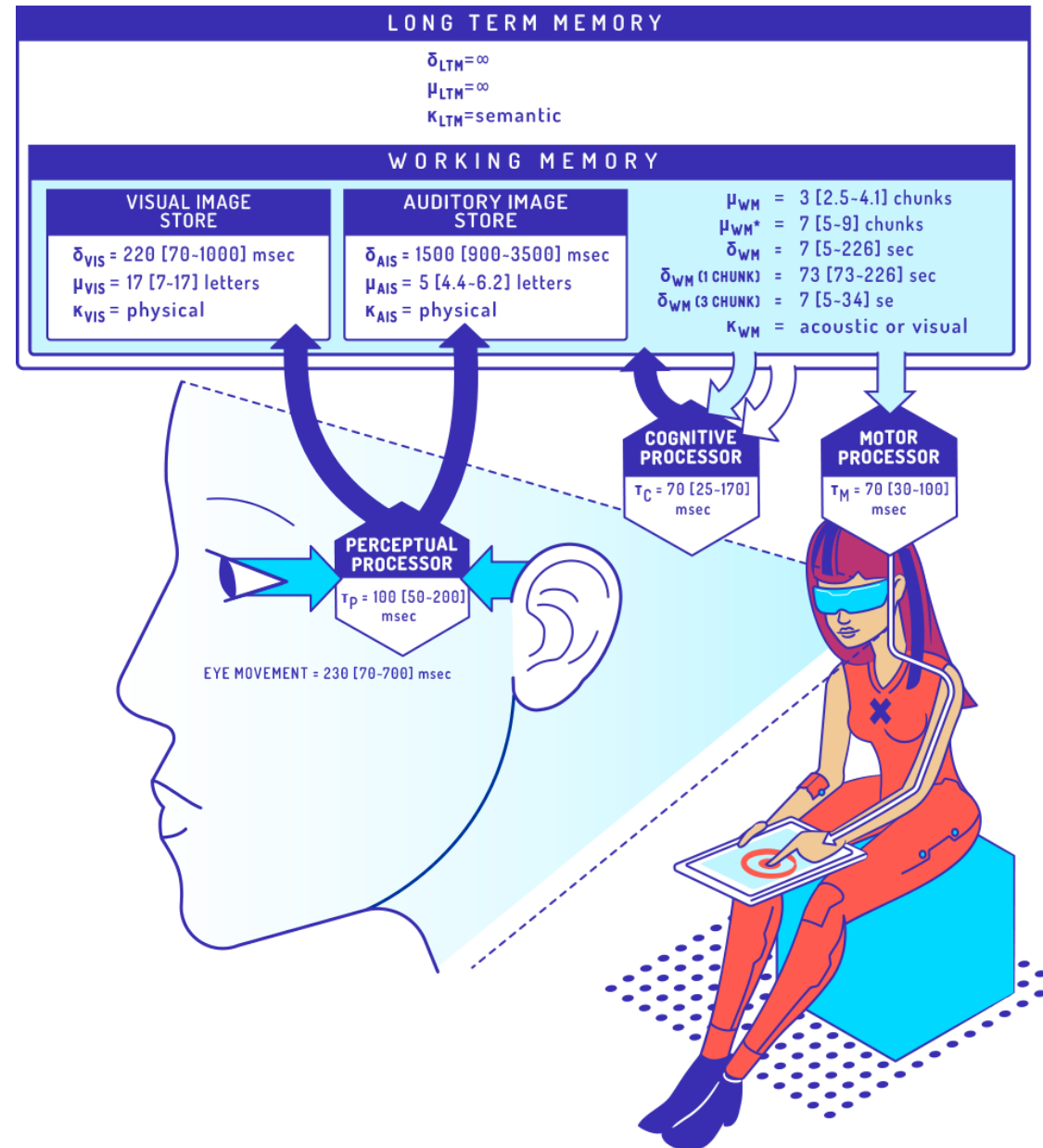
Stimuli ■————→ Behaviour

MODEL HUMAN PROCESSOR

GRAPHICS PROGRAMMING

UNIT 1: HUMAN PERCEPTION

INTRO



DEFINITION

Perception is the processing of information from environmental stimuli (external perception) or internal bodily signals (interoception).

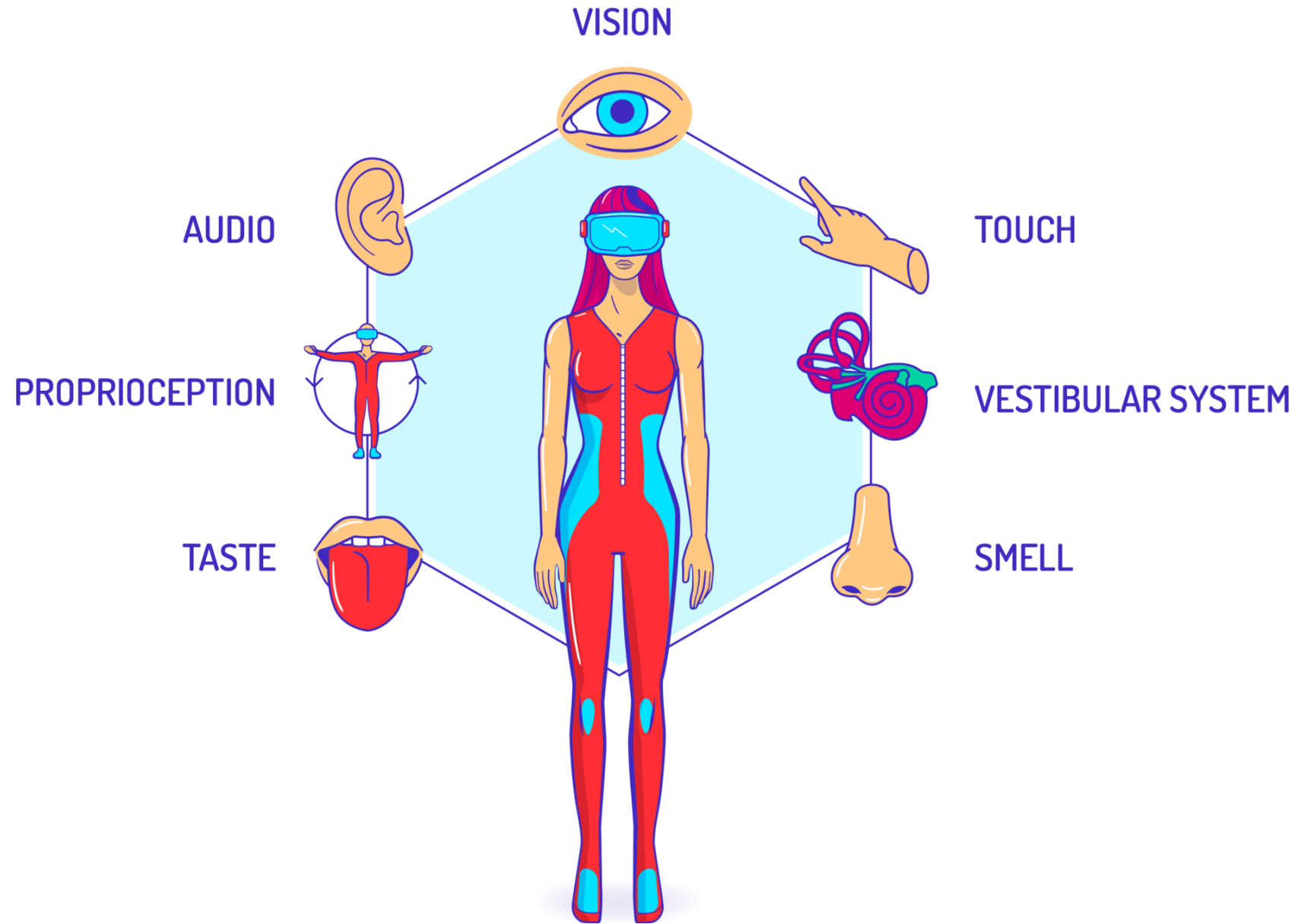
Filtering, selection and integration of information is part of perception.

SENSORY MODALITIES

GRAPHICS PROGRAMMING

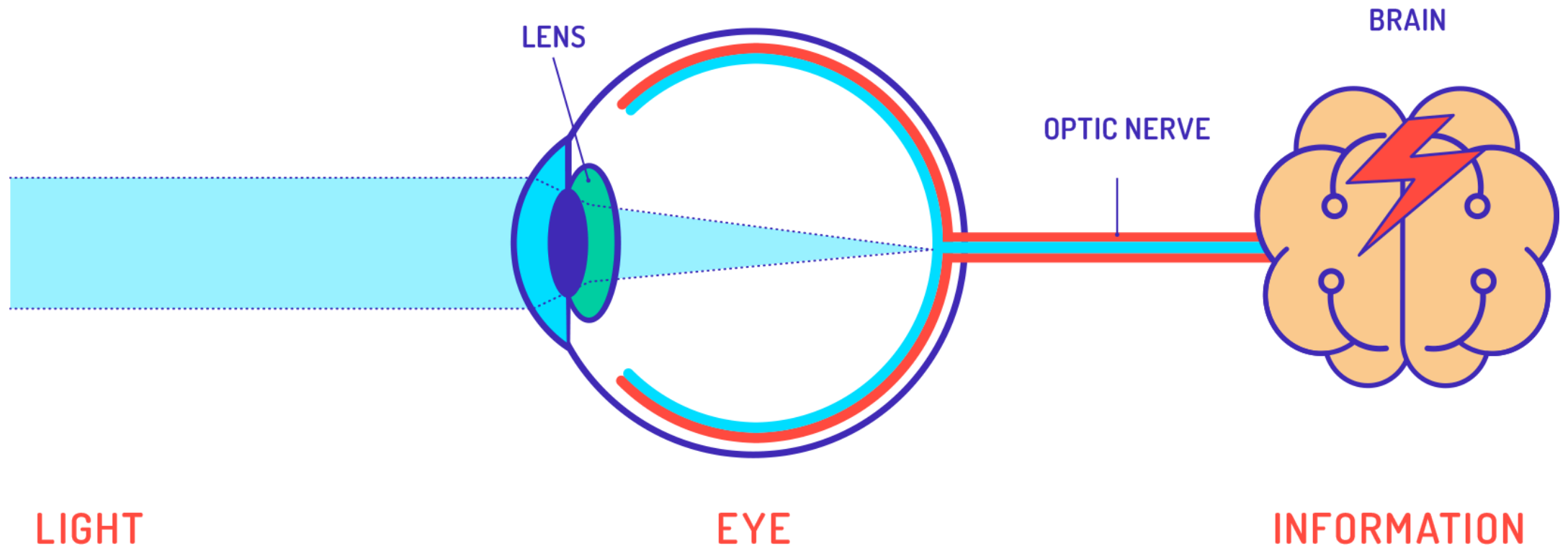
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PERCEPTION



VISION

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UNIT 1: HUMAN PERCEPTION
VISUAL PERCEPTION

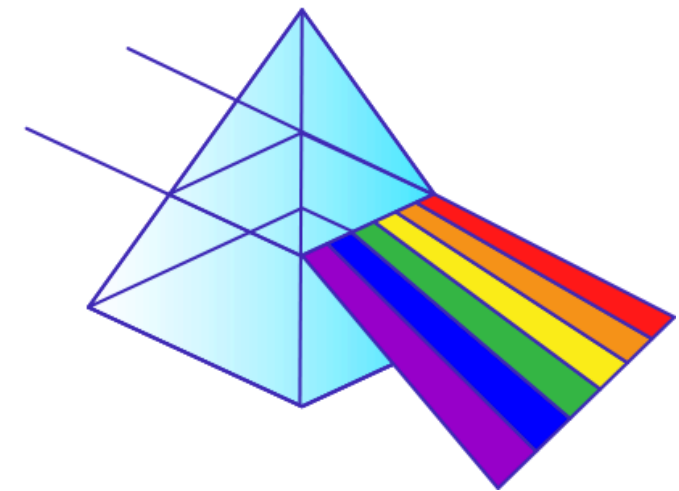
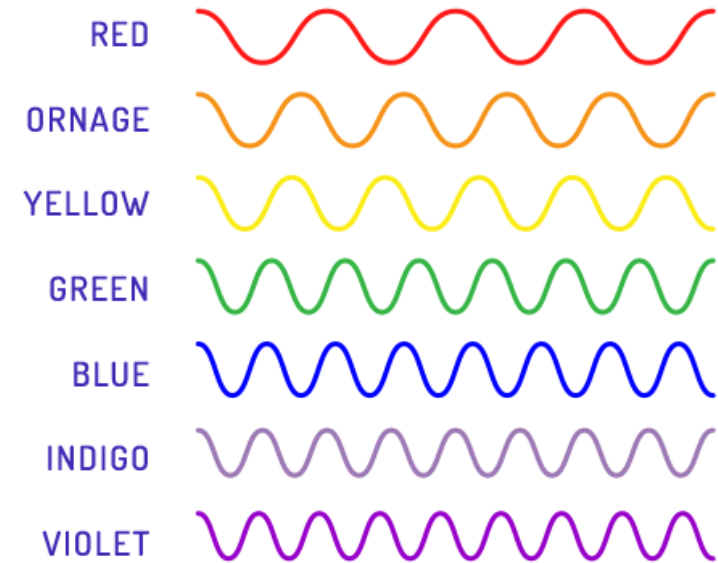
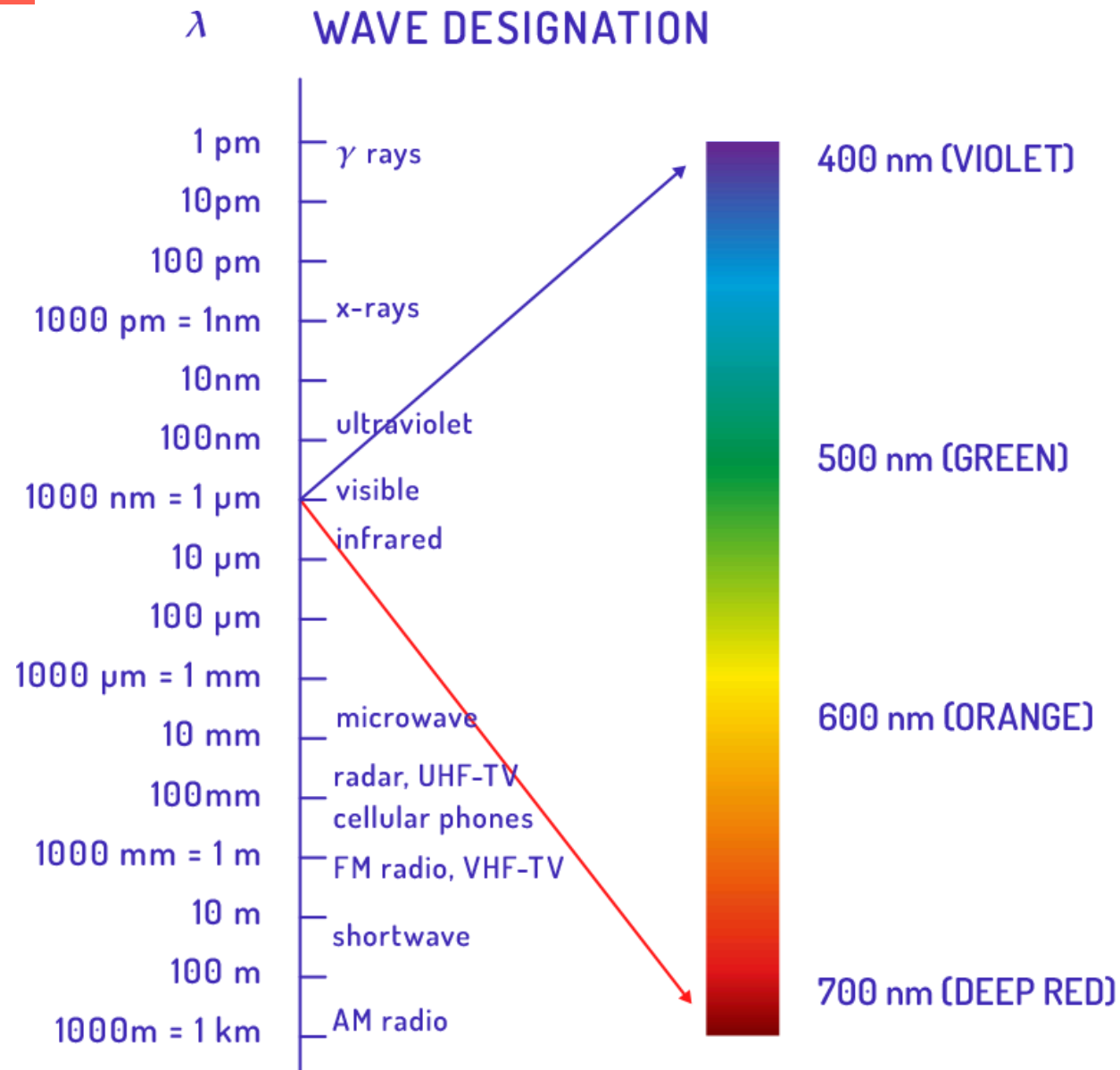


LIGHT

GRAPHICS PROGRAMMING

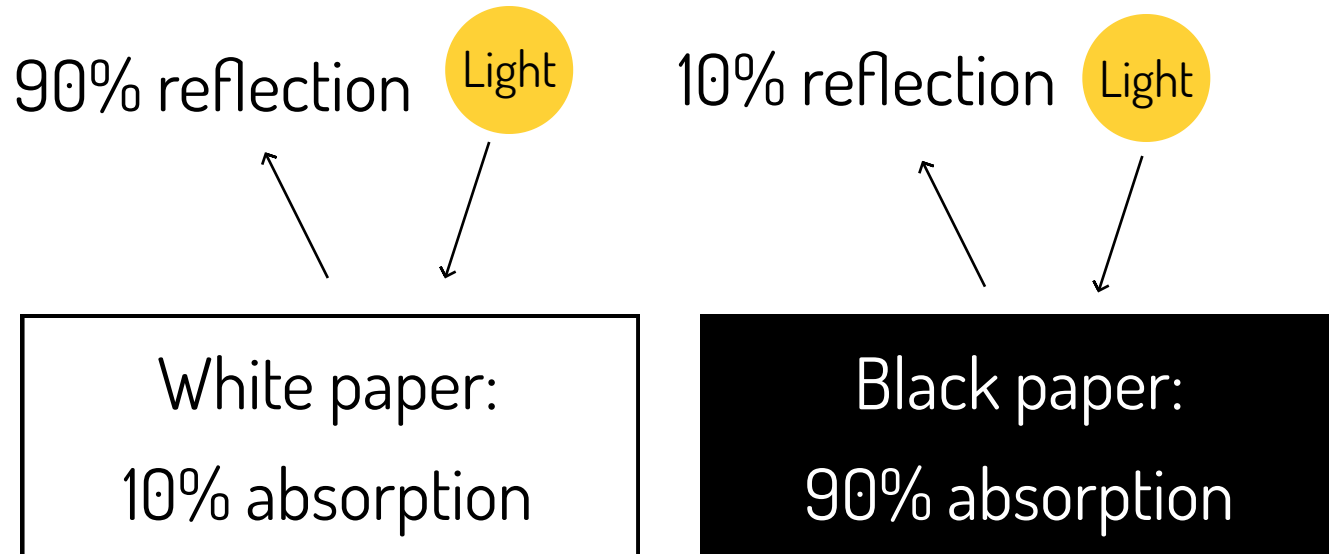
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VISUAL PERCEPTION



LIGHT

- Light is electromagnetic wave
 - frequency defines color
 - amplitude defines brightness
- Light is reflected on objects
- Light enters **directly** or **indirectly** our eyes

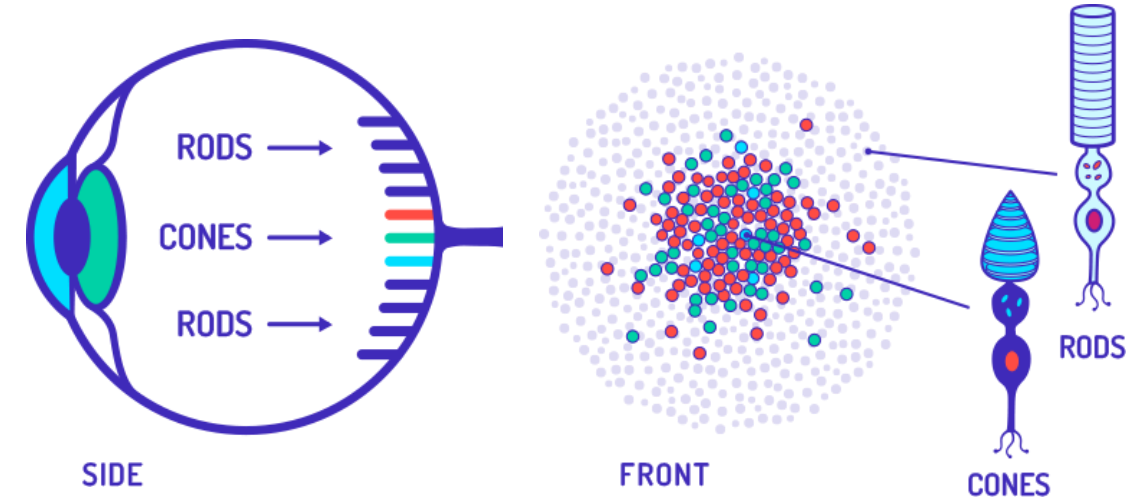
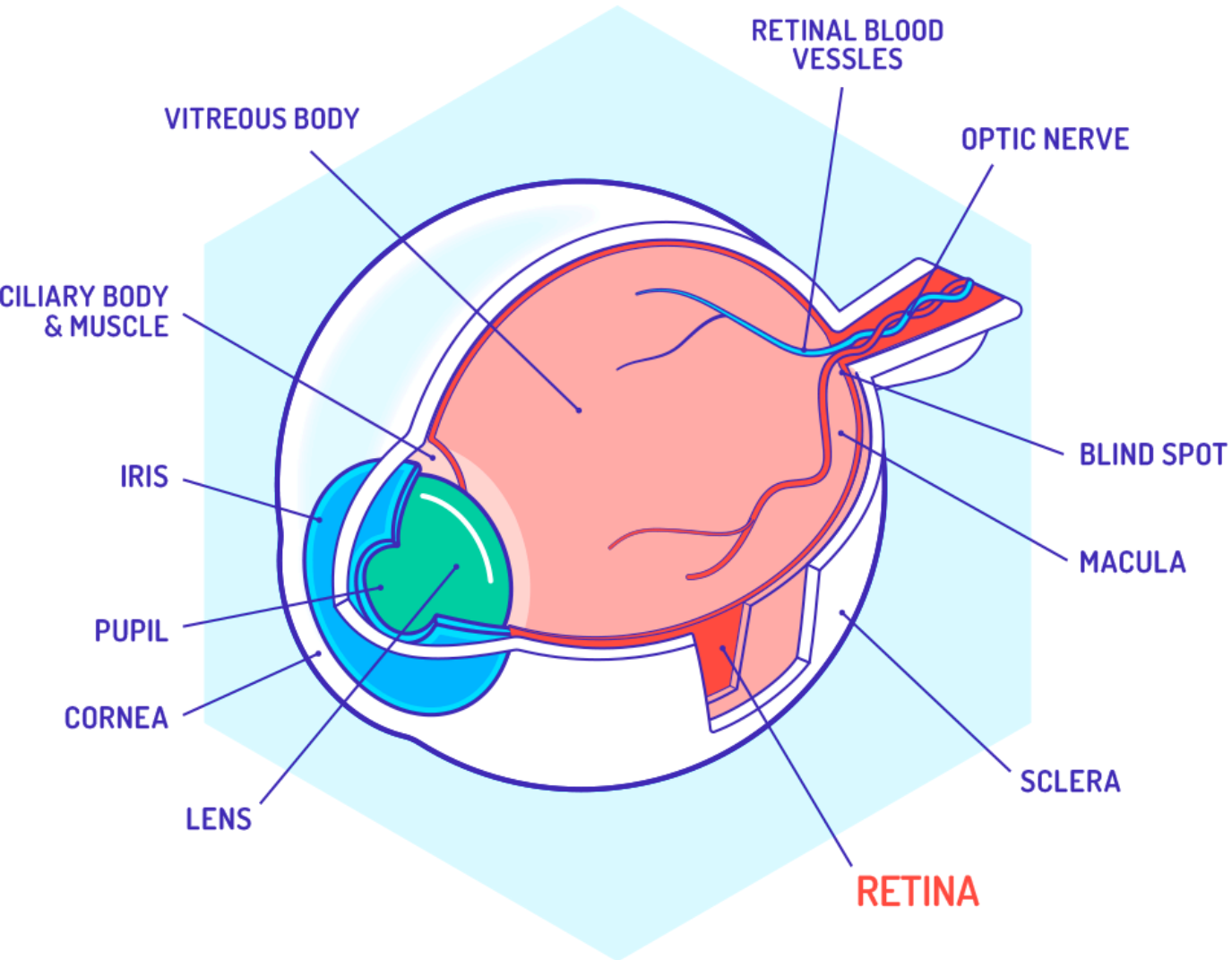


HUMAN EYE

GRAPHICS PROGRAMMING

UNIT 1: HUMAN PERCEPTION

VISUAL PERCEPTION



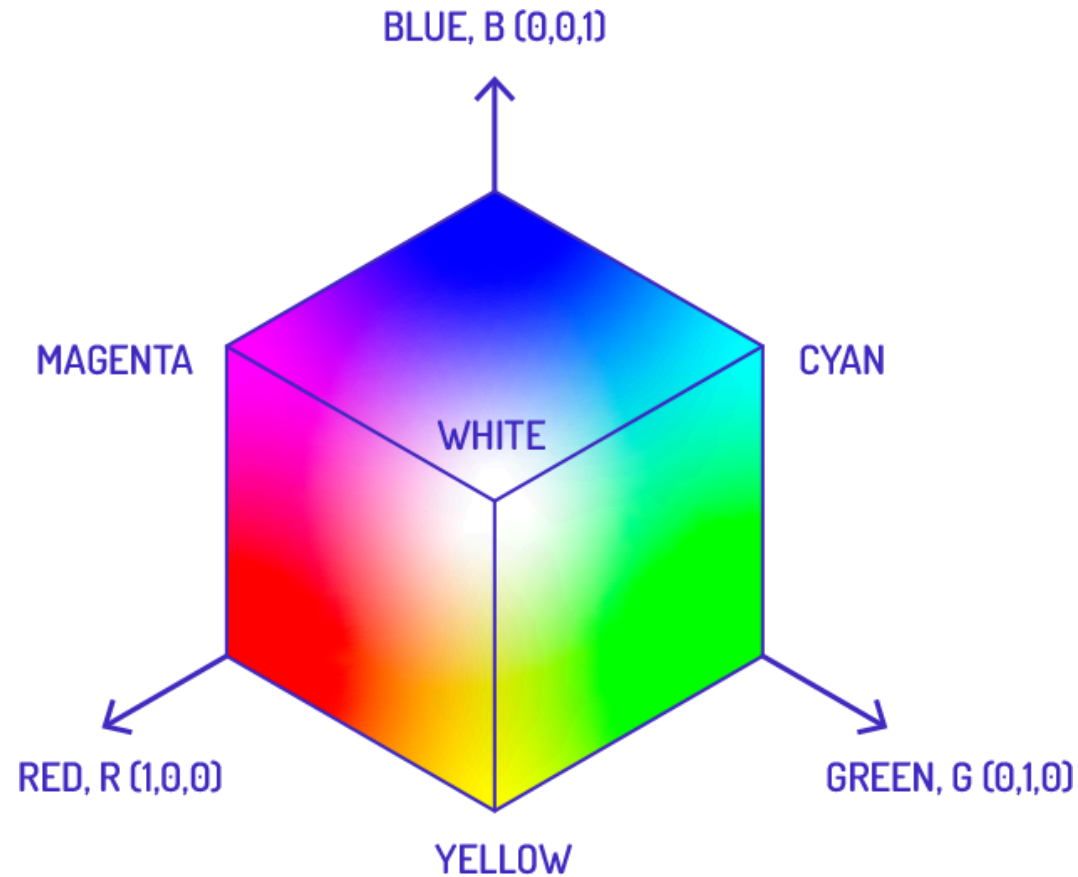
- Perception of color, shape, depth, movements etc.
- **RETINA** is innermost, light-sensitive layer of the eye
- Photoreceptive cells transform light into neuronal activities and forward these information
 - **CONES (7-8 MIO)**: colors & shape
 - **RODS (100-120 MIO)**: light/dark, motion

COLOR SPACE

GRAPHICS PROGRAMMING

UNIT 1: HUMAN PERCEPTION

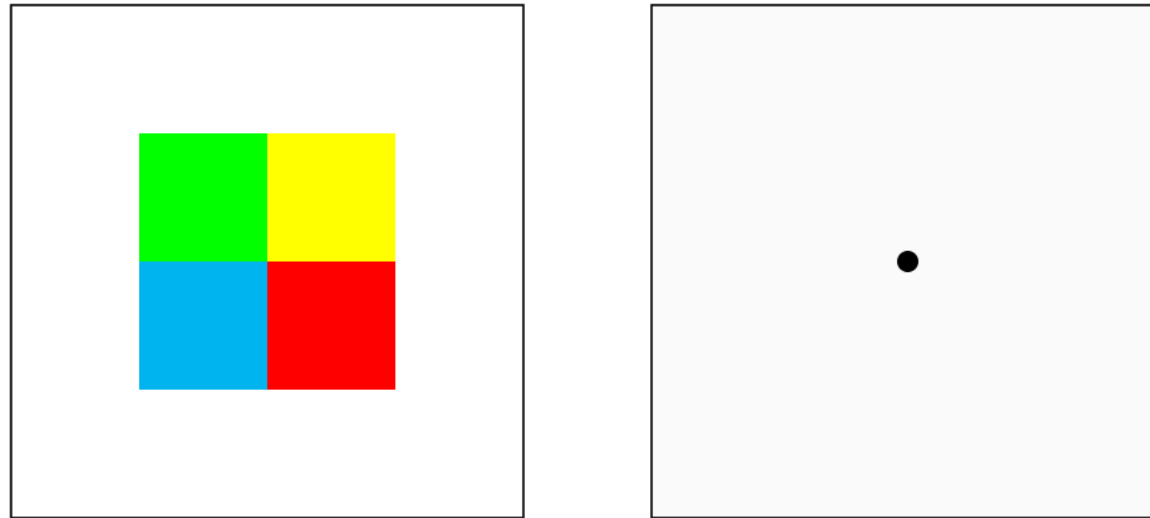
VISUAL PERCEPTION



RGB COLOR SPACE

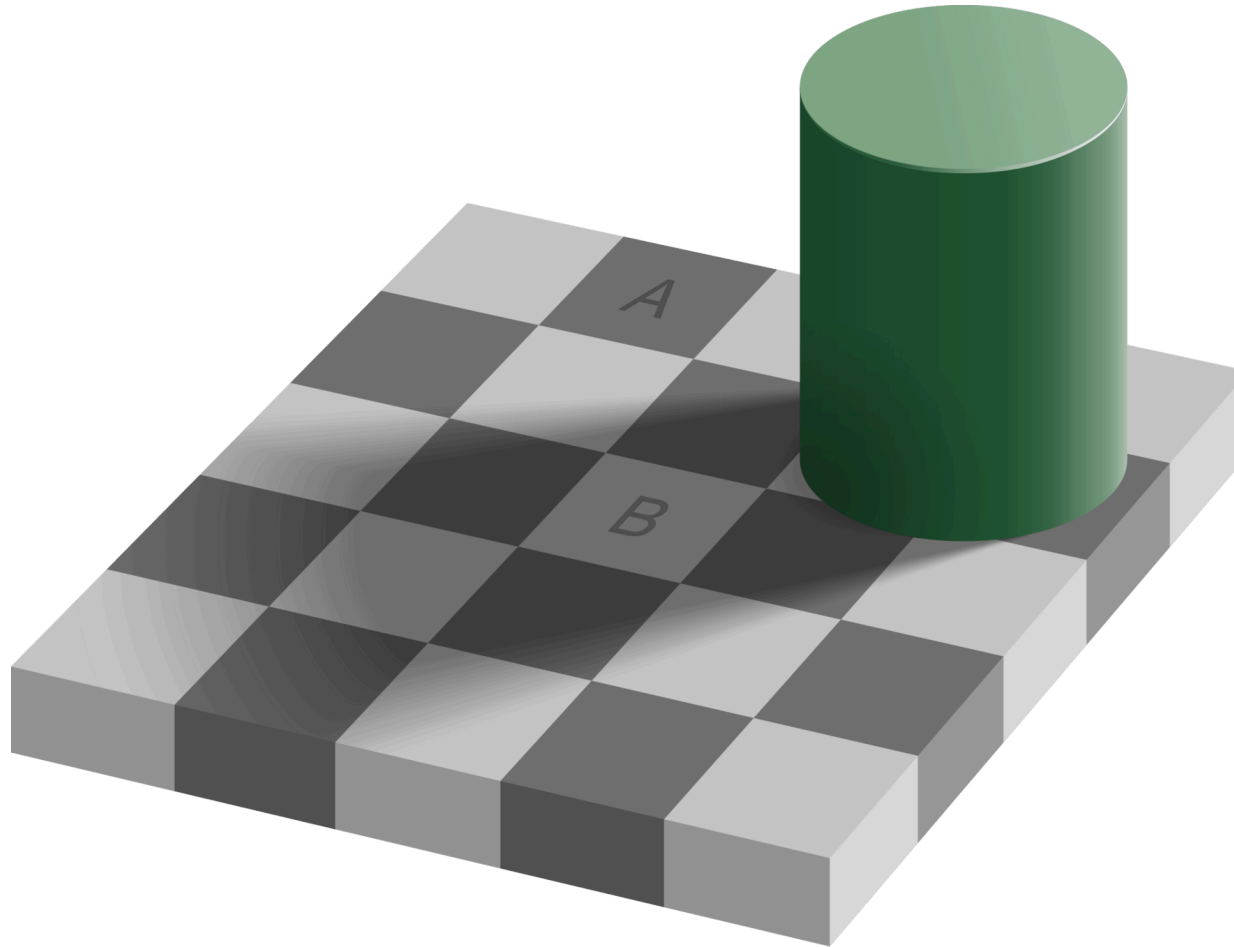
OPPONENT-PROCESS THEORY

- **Mechanism:** Antagonistic pairings in the visual system.
- **Pairs:** Red-Green, Blue-Yellow, and Black-White.
- **Evidence:** Explains afterimages (staring at green leaves creates a red ghost image).



COLOR CONSTANCY

- The brain's ability to perceive consistent color despite changing illumination.



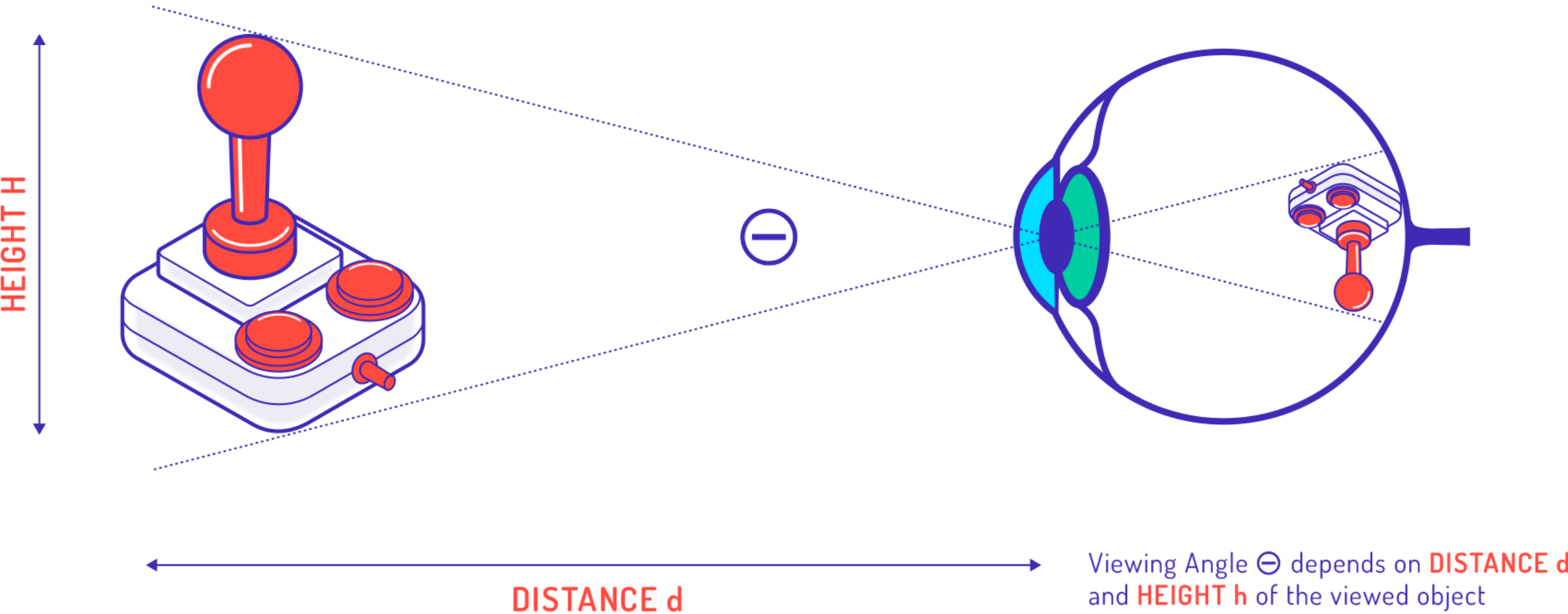
Source: Wikipedia

SIZE PERCEPTION

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VISUAL PERCEPTION

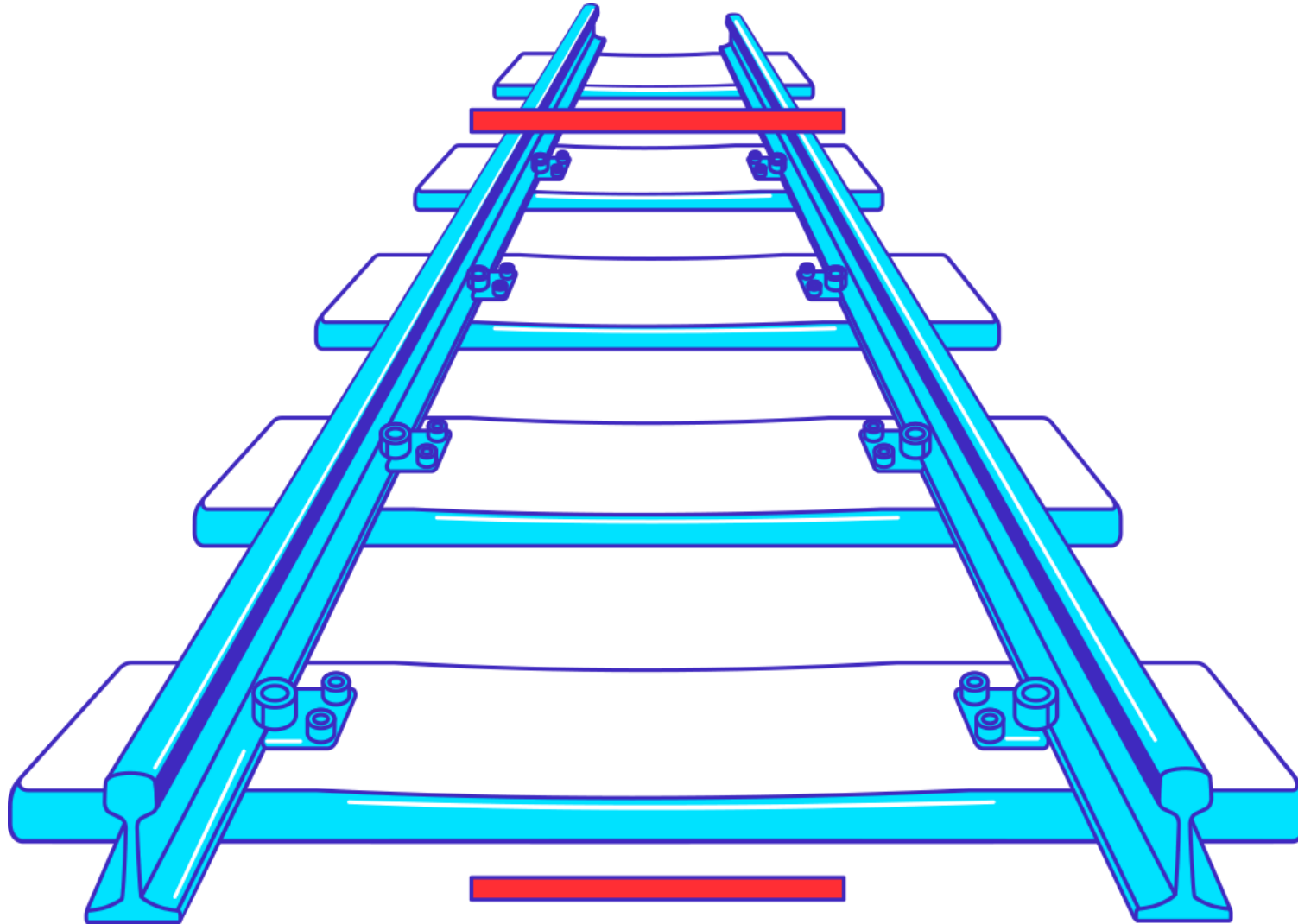


PONZO ILLUSION

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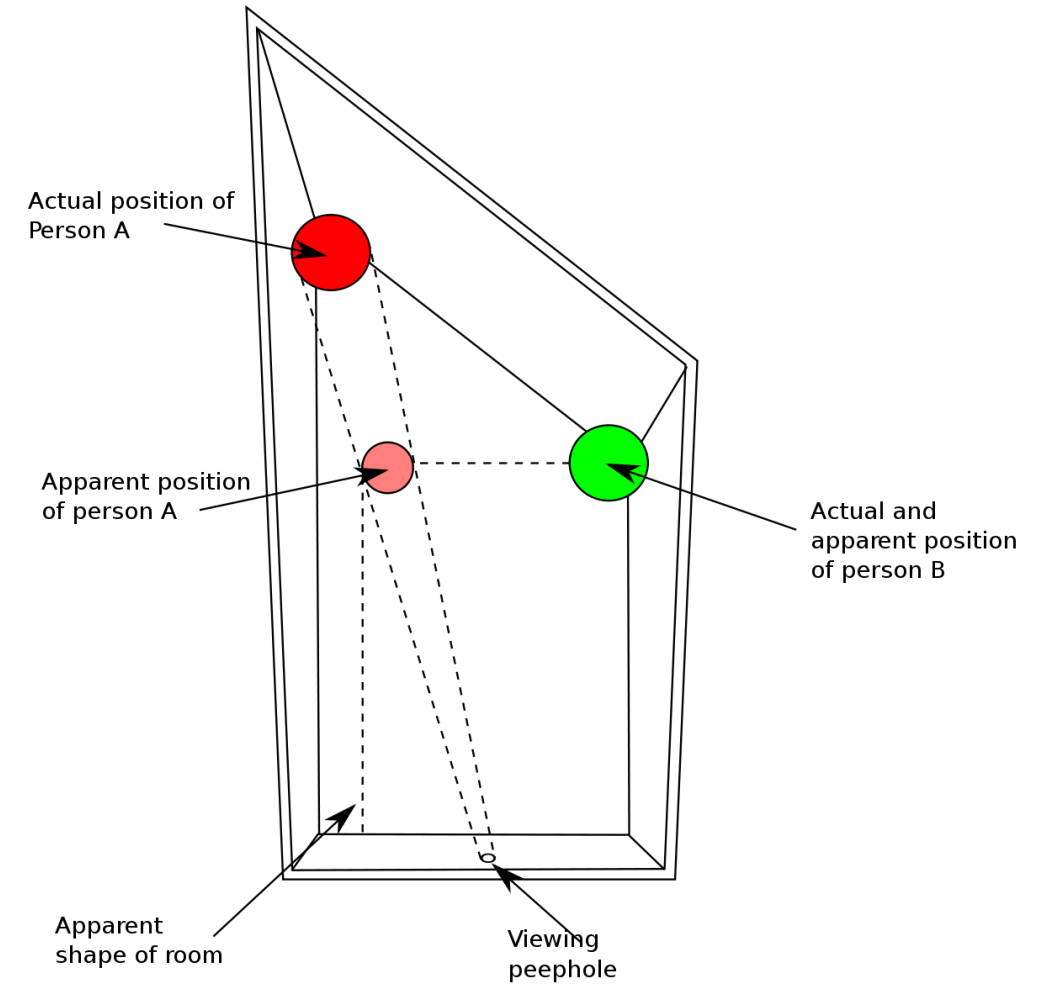
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VISUAL PERCEPTION



AMES ROOM

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UNIT 1: HUMAN PERCEPTION
VISUAL PERCEPTION

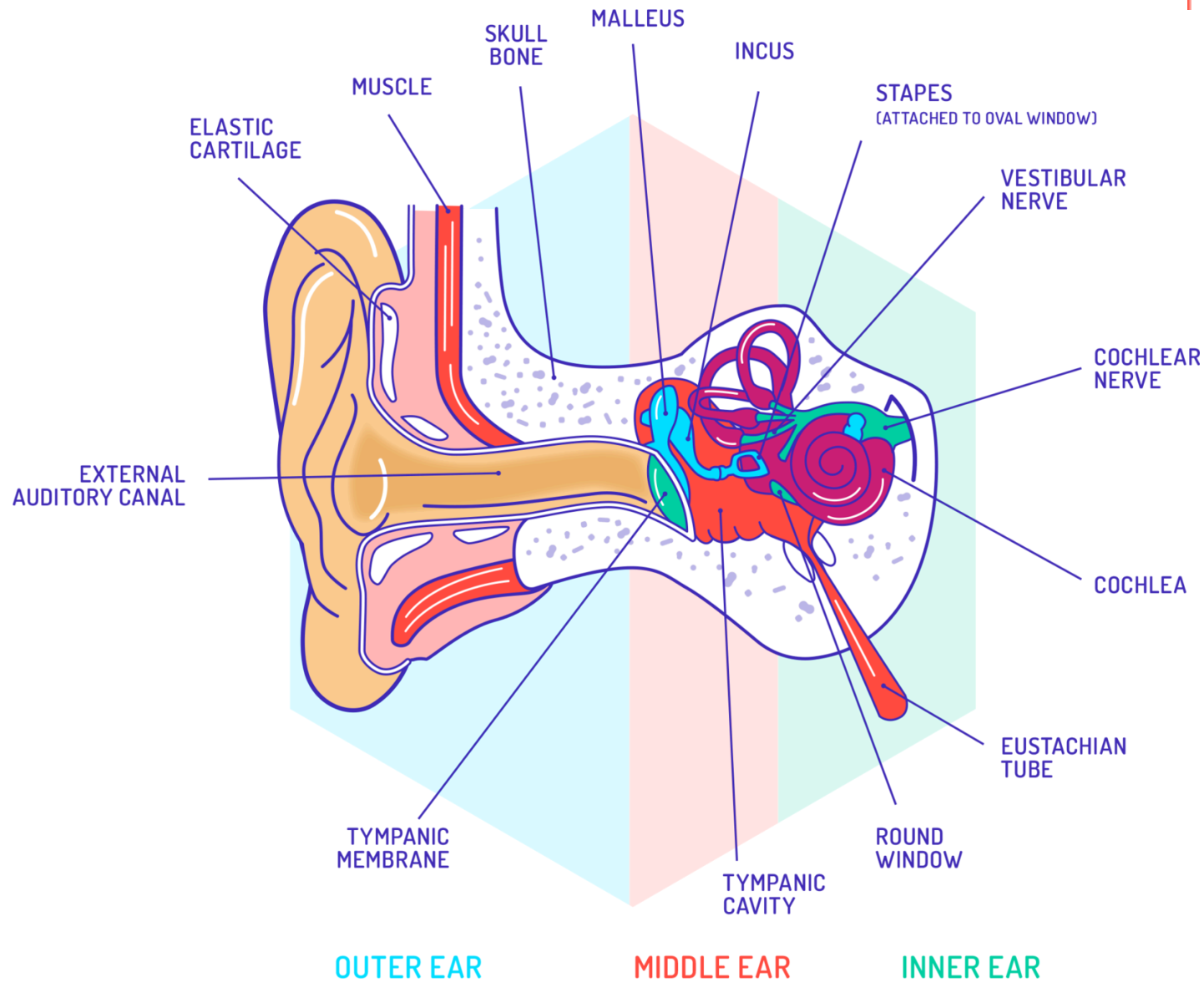


AUDIO

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SENSORY CHANNELS

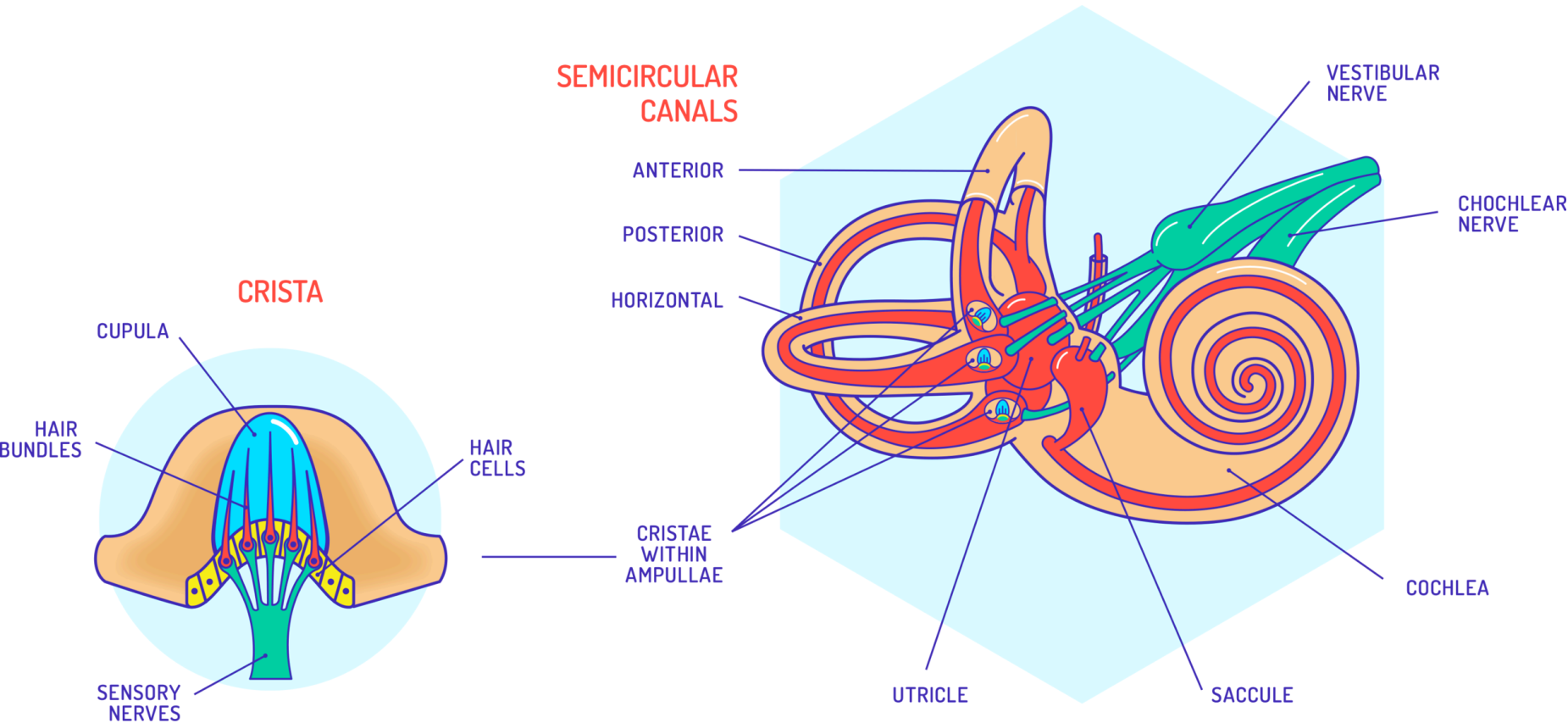


VESTIBULAR SYSTEM

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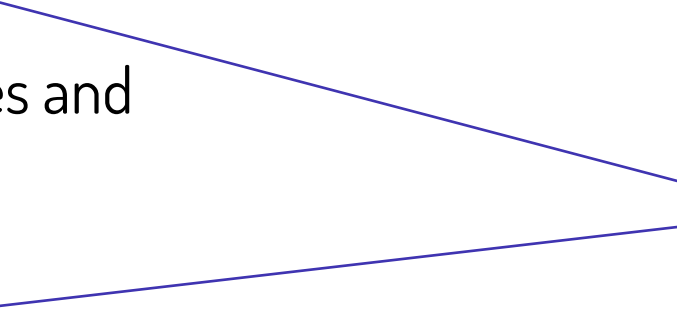
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SENSORY CHANNELS



- Vestibular channel is able to perceive linear and angular accelerations
- Otolith organs (saccule and utricle): gravity and linear movements
- Semicircular canals: rotational movements
- Adaption to constant stimulation
- Otolith organs can not distinguish between a head tilt and linear movement since they perceive only the overall sum of gravity and inertial acceleration (GIA vector)

TOUCH

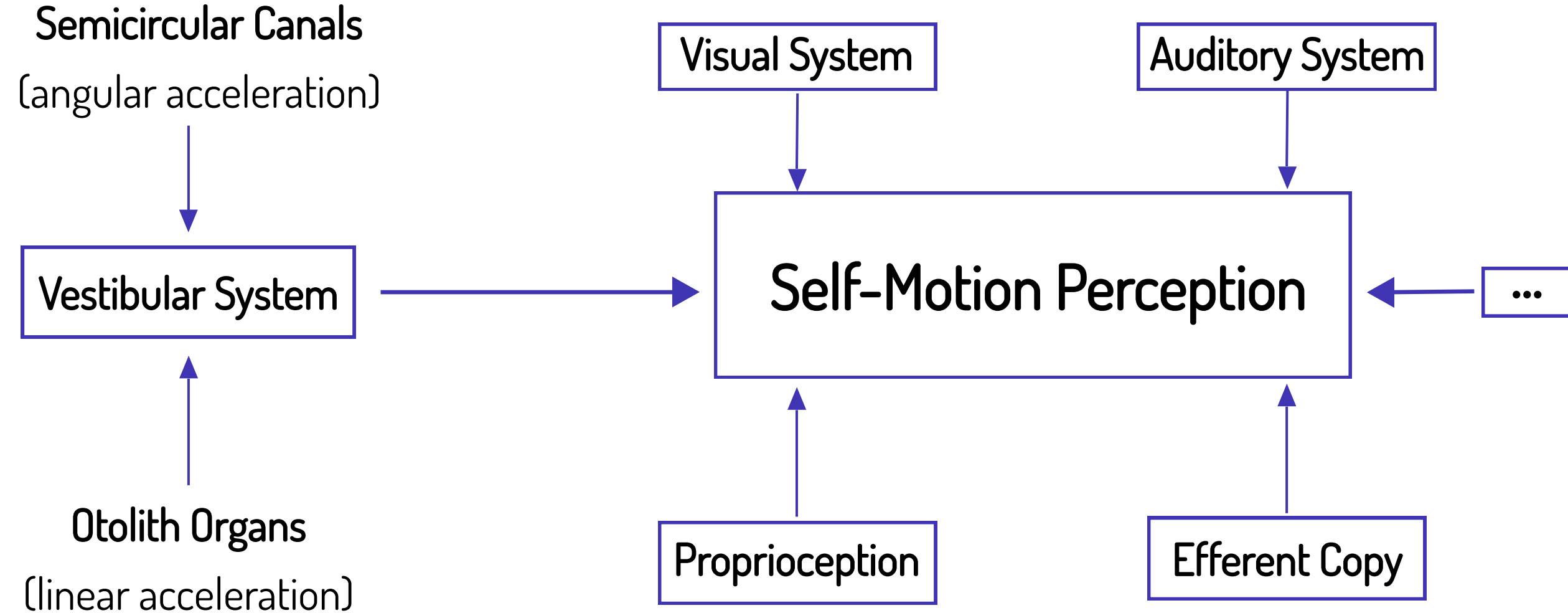
- Haptic perception
 - receptors of muscles and joints
 - Tactile perception
 - receptors under the skin
 - pressure, pain and temperature
 - Proprioception
 - perception of self-movements (kinesthetics)
 - position and orientation of the body and limbs
- passive or active
- 

MULTI-SENSORY INTEGRATION

GRAPHICS PROGRAMMING

UNIT 1: HUMAN PERCEPTION

SENSORY INTEGRATION



MULTI-SENSORY INTEGRATION

GRAPHICS PROGRAMMING

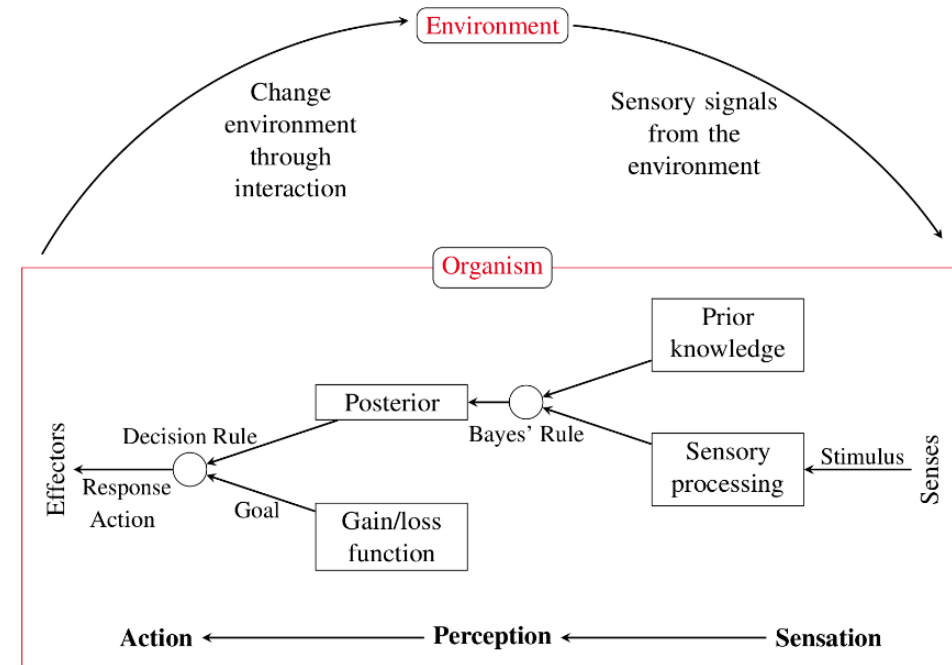
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SENSORY INTEGRATION

- Maximum Likelihood Estimation:
 - Sum of individual information from all sensory channels
 - Each channel different weight depending on quality
 - vision tends to dominate the other sensory channels in many situations

$$I = \sum_{i=1}^N w_i I_i \text{ with } \sum_{i=1}^N w_i = 1$$

- Bayes' Rule:
 - describes the combination of information from sensory channels and prior knowledge
 - Each channel different weight depending on quality
 - vision tends to dominate the other sensory channels in many situations



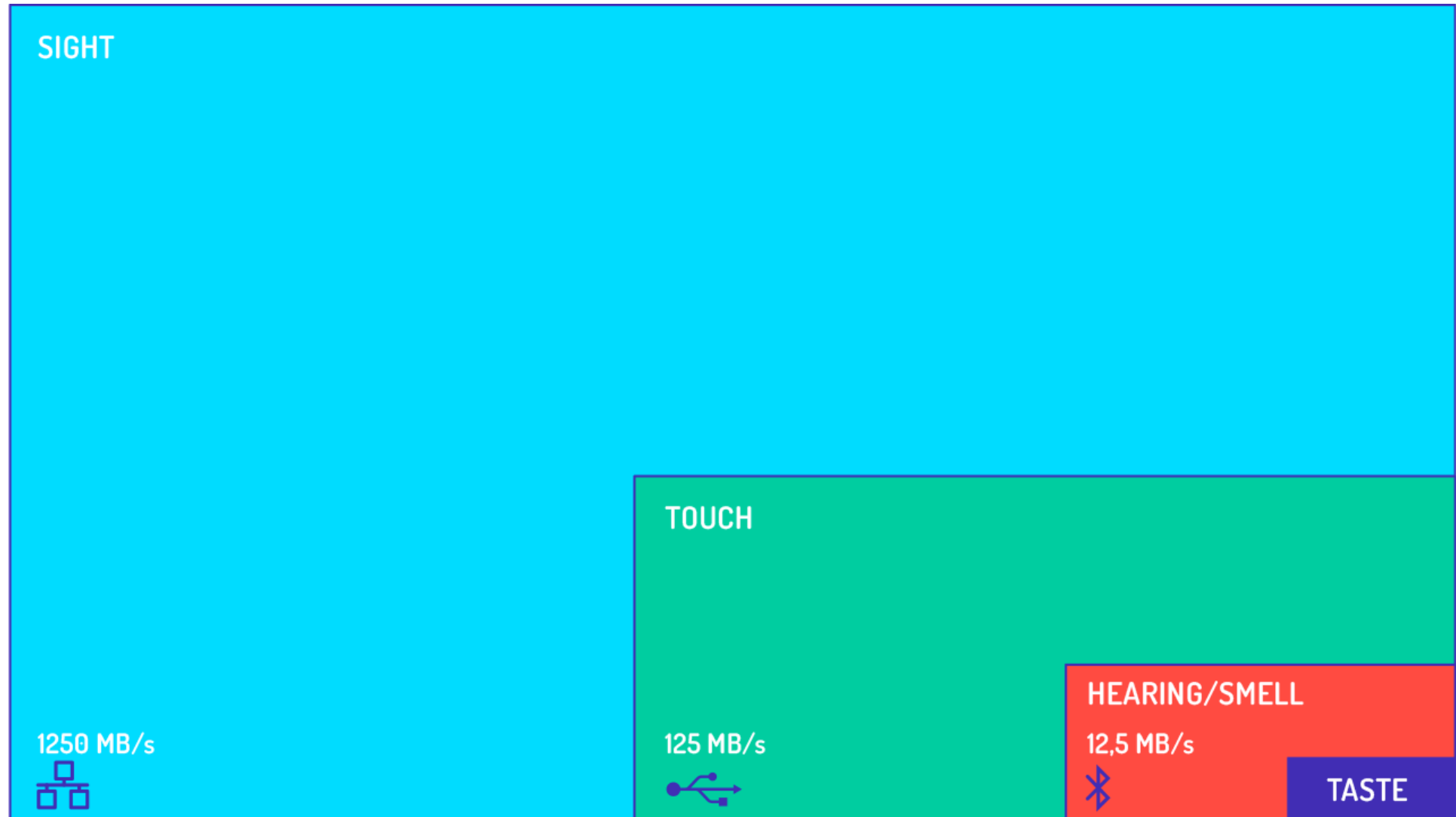
$$p(S|O) = \frac{p(O|S) \times p(O)}{p(O)}$$

HUMAN BANDWIDTH

GRAPHICS PROGRAMMING

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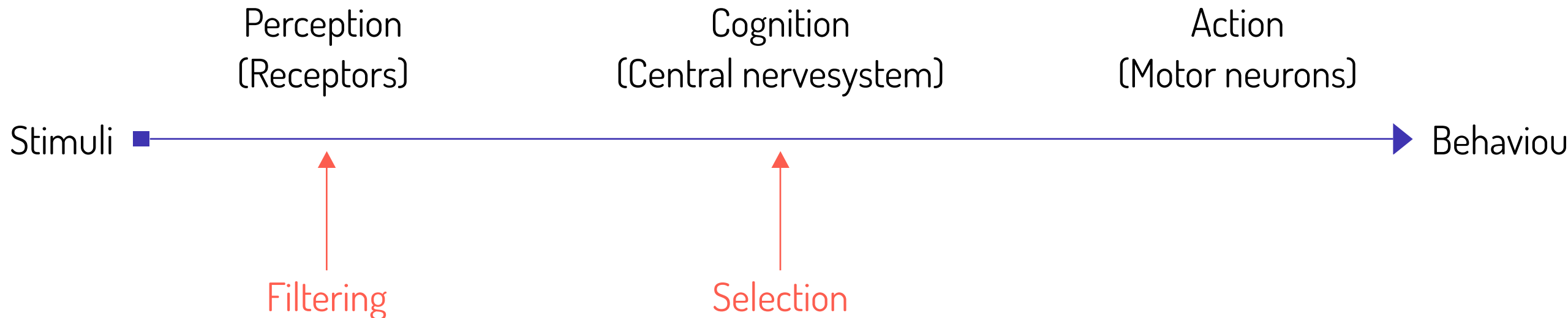
SENSORY INTEGRATION



D. McCandless: The beauty of data visualization, TED Talk, 2010

FILTERING & SELECTION

- Limited capacity for storage and transformation
 - perception
 - memory
- inter- and intra-individual differences
- Total amount of input information: 1.5 GBit/s
 - ca. 15 MBit/s reach receptors
 - ca. 100 MBits/s reach consciousness



SELECTIVE PERCEPTION

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UNIT 1: HUMAN PERCEPTION
SELECTION



Video: No CC-License

CROSS-MODAL PERCEPTION

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UNIT 1: HUMAN PERCEPTION
SELECTION



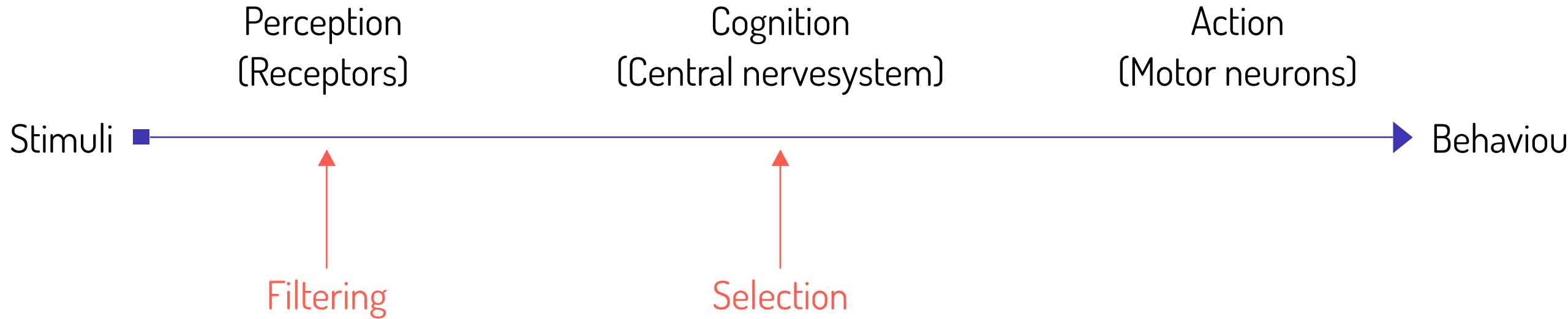
H. McGurk et al.: Hearing Lips and seeing voices, Nature, 1976 (No CC-License)

COGNITION

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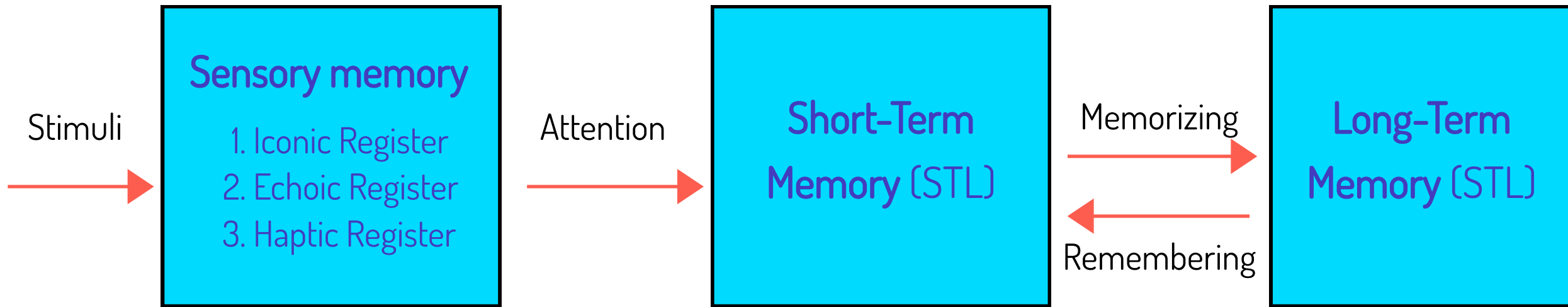
COGNITION



- **Storage:** Memory & Remembering
 - Sensory Store
 - Working memory
 - Long-term memory
- **Attention**
- **Transformation:**
 - Logic
 - Problem Solving
 - Learning/Training
- **Decision making**
- **Cognitive load**

MULTI-STORAGE MODEL

GRAPHICS PROGRAMMING
UNIT 1: HUMAN PERCEPTION
COGNITION



R.C. Atkinson, R.M. Shiffrin: Human memory: A proposed system and its control processes.
The psychology of learning and motivation (Volume 2). pp.89-195, 1968.

QUESTIONS?

UNIT 1: HUMAN PERCEPTION



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